

Coding & Solution

Date	11 july 2026
Team ID	xxxxxx
Project Name	EV Data Analytics & Visualization Dashboard
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/phani-cs/Ev-dataanalytics
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	Flask, Tableau Desktop, Tableau Public, Bootstrap, Pandas
Key Features Implemented	EV Charging Pattern Analysis Dashboard, Battery Performance Dashboard, EV Model Comparison Dashboard, Interactive Filters, KPI Cards, Calculated Fields, Data Preprocessing, MySQL Database Integration, Tableau Story, Flask Web Integration.
Pending / Incomplete Features	Real-time EV Charging API Integration, Live Battery Monitoring, Predictive Analytics, AI-based Demand Forecasting, User Authentication
Setup / Run Instructions	Clone the repository, install Python dependencies (pip install -r requirements.txt), configure the MySQL database, import the EV dataset, connect Tableau to MySQL, publish the dashboards, run the Flask application (python app.py), and access the dashboard through the local browser.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The **EV Data Analytics & Visualization Dashboard** follows software engineering best practices by maintaining a modular architecture consisting of data preprocessing, database management, Tableau dashboards, and Flask web integration. The project uses **Python (Pandas)** for data preparation, **MySQL** for data storage, **Tableau** for interactive visualizations, and **Flask** to embed dashboards into a web application. The codebase is maintained using **Git and GitHub**, enabling version control and collaborative development. The architecture is scalable and allows future enhancements such as **real-time EV charging data integration, AI-based demand forecasting, battery health prediction, and predictive analytics**.